How Much Do Time-Series Anomaly Detection Metrics Actually Disagree?
Null Models and Cluster-Aware Inference for Rank-Flip Statistics

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Abstract

Reports that point-level and segment-level metrics rank time-series anomaly detectors differently have become a standard argument for evaluation reform. These reports take the form of a *rank-flip rate*: the fraction of model pairs that two metrics order differently. We audit that statistic on a 180-series, 25-model recomputation of TSB-AD-M spanning 17 source collections, and on a 6-dataset, seven-detector grid. Our main finding is that evaluation series are treated as independent when they are not, and that this invalidates the standard α-based structural explanation built on top of the statistic. The covariate most often invoked to explain disagreement, the short-anomaly ratio α, is constant within 11 of the 12 collections that contain more than one series: it is a collection label rather than a series-level variable. Its rank correlation with the flip rate falls from 0.3241 at the series level to 0.0563 ($p = 0.827$) with collections as the unit, and to 0.0898 when one collection is removed. Interval estimates move the same way, from $[0.2979, 0.3312]$ resampling series to $[0.2695, 0.3715]$ clustering over collections; on the six-dataset grid the clustered interval spans $[0.0667, 0.4167]$, which cannot resolve its magnitude. Two further omissions point the same way. Raw flip percentages are reported without stating the chance level, which is 0.5 and not 0: our observed 0.3145 against a permutation null of 0.5002 means the metrics agree on 68.6% of pairs. And flips are pooled across margins: among the 13.3% of pairs that both metrics separate by at least 0.20, a fraction of 0.0220 are ordered differently, so confident disagreement is roughly two percent rather than the 0.3145 that gets reported. We apply the audit to a composite metric we previously proposed ourselves and show it is uninformative by construction at both ends of the regime it was built to span. Applying the same corrections to our own positive findings leaves one covariate standing, segment count, at -0.6225 ($p = 0.008$) with collections as the unit. We give a reporting protocol and release code that regenerates every number here from committed artifacts.

1 Introduction

Time-series anomaly detection (TSAD) evaluation has been under sustained criticism for a decade. Point-adjustment inflates scores to the point where random detectors appear competitive (Kim et al., 2022); widely used benchmarks contain mislabeled, trivially separable, or unrepresentative anomalies (Wu & Keogh, 2023; Liu & Paparrizos, 2024); and a series of replacement metrics has been proposed to repair specific failures of point-wise scoring (Tatbul et al., 2018; Huet et al., 2022; Paparrizos et al., 2022; Ghorbani et al., 2024).

A particular kind of evidence recurs throughout this literature. One computes two metrics over the same detector–dataset grid, counts the model pairs that the two metrics order differently, and reports the fraction.

[1]Large language models were used as assistive tools in the statistical re-analysis and in drafting this manuscript. All reported numbers are produced by the released scripts and were verified against the committed artifacts by the author, who takes full responsibility for the content.

We call this a *rank-flip rate*. It is rhetorically powerful: a statement that a third of published rankings would reverse under a defensible alternative metric implies that leaderboard positions carry little information. We used exactly this argument in earlier work.

This paper asks what a rank-flip rate actually measures. Our finding is that the statistic as reported is missing three ingredients, that each omission pushes the interpretation in the same direction—toward overstating instability—and that supplying them changes several conclusions in this literature, including our own.

A stated chance level. A flip rate is a fraction between 0 and 1, and it is discussed as though 0 were the reference point, so that any positive value reads as evidence of disagreement. The correct reference is the value the statistic takes when the two metrics share no information about model quality, which is 0.5. We are careful about the scope of this claim. The field does baseline against chance elsewhere: Yang et al. (2025) evaluate metric *values* under random detection, and Lyu (2026) reports inter-metric Kendall's τ, which is itself chance-corrected. What is missing is narrower and specific: raw flip *percentages* are reported and interpreted without the chance level being stated alongside them. The move of supplying a permutation null for pairwise reversal rates has also been made independently and concurrently outside our field—Kendiukhov (2026) does exactly this for gene regulatory network benchmarking, obtaining a null of 0.500 against an observed 0.163—which we take as corroboration rather than as a contribution of ours.

A margin treatment. Whether two metrics order a pair of models differently is only meaningful if the models are actually distinguishable. Pairs separated by a negligible margin flip at close to the chance rate by construction, and they make up a substantial share of every grid. The margin must moreover be taken on both metrics: conditioning only on the primary one selects pairs it separates confidently while leaving the secondary margin free, which is exactly where the flips turn out to be.

A cluster unit. Evaluation series are routinely treated as independent replicates. They are not: benchmark suites are assembled from source collections, and series within a collection share a sensor layout, a labeling convention, and an anomaly-generating process. This matters most for exactly the covariates used to *explain* disagreement, because those covariates tend to be properties of the collection rather than of the series. When that is the case, a structural explanation supported at $n = 180$ can be supported by, in effect, a handful of independent observations.

Contributions. Our primary contribution is the third item below; the first two are corrections that are individually straightforward and, in the case of the null, partly concurrent with work in another field.

1. We supply all three ingredients on two grids: a 180-series, 25-model, 4,498-row recomputation of TSB-AD-M spanning 17 source collections, and a 6-dataset, seven-detector study on TAB-style benchmarks (Section 2).
2. We show that reported flip rates describe agreement once the chance level is stated, and that most of the residual is a margin artifact that survives only under one-sided conditioning (Sections 3–4).
3. **We show that structure-conditioned explanations of disagreement are confounded with collection identity**, and that the standard thresholded short-anomaly ratio is not identifiable at the series level at all. We are not aware of a precedent for this analysis of rank-flip statistics, in TSAD or elsewhere (Section 5).
4. We give a worked failure case: a metric defined as an interpolation between two others is uninformative at both endpoints of the very regime it was introduced to span. The metric is one we proposed ourselves, and the analysis we offered in support of it cannot come out any other way (Section 6).
5. We identify two continuous covariates that do survive clustering, and give a reporting protocol (Sections 7–8).

We are not arguing that TSAD evaluation is sound. Two of the criticisms above survive our audit intact, and a third—that benchmark anomalies are structurally unlike the point-wise events that point metrics score—we confirm as a factual matter. We are arguing that the specific statistic most often used to *quantify* the problem does not support the weight placed on it, and that this is fixable.

2 Setup

Grids. We use two evaluation grids. The first is a 6-dataset study on TAB-style benchmarks (Qiu et al., 2025): SWaT (Mathur & Tippenhauer, 2016; Goh et al., 2017), MSL and SMAP (Hundman et al., 2018), WADI (Ahmed et al., 2017), PSM (Abdulaal et al., 2021), and SMD (Su et al., 2019), evaluated with five deep detectors—Anomaly Transformer (Xu et al., 2022), TranAD (Tuli et al., 2022), CATCH (Wu et al., 2025), DADA (Shentu et al., 2025), and MOMENT (Goswami et al., 2024)—plus two classical baselines, Isolation Forest (Liu et al., 2008) and LOF (Breunig et al., 2000). The second is a replication on the TSB-AD-M evaluation list (Liu & Paparrizos, 2024), covering 180 multivariate series and 25 scored models for 4,498 model–series metric rows, drawn from 17 source collections.

Metrics. We report AUC-ROC and segment-level affiliation F1 (Huet et al., 2022) computed with `prts` at cardinality `one` and flat bias, with thresholds set from the contamination rate of the test partition following TAB's protocol (Qiu et al., 2025). Contamination-rate thresholding is itself an assumption with known sensitivity (Masakuna et al., 2024; Perini et al., 2023); our analysis conditions on it throughout and we do not claim results are invariant to it.

Rank-flip rate. For a group g (a dataset, or a series) with model scores a_i under a primary metric and b_i under a secondary metric, we count unordered pairs (i, j) with $a_i \neq a_j$ and $b_i \neq b_j$, and call a pair *flipped* when $\text{sign}(a_i - a_j) \neq \text{sign}(b_i - b_j)$. The flip rate pools flipped pairs over comparable pairs. This matches the definition used in our own earlier analysis, which is the instance we audit directly here.

Anomaly-duration taxonomy. Anomaly segments are contiguous runs of positive labels in the test partition, bucketed as Short (≤ 5 steps), Medium (6–50), or Long (> 50). The structure weight in common use is $\alpha = 1 - N_{\text{short}}/N_{\text{total}}$. On the 6-dataset grid, only PSM ($\alpha = 0.4861$) and SMD ($\alpha = 0.6147$) have $\alpha < 1$; SWaT, MSL, SMAP, and WADI have no Short segments at all. That last fact is a property of the labels and is not disputed here.

Reproducibility. Every number in this paper is emitted as a LaTeX macro by a script that reads the committed JSON artifacts, so the manuscript cannot drift from the analysis. Rerunning three analysis scripts regenerates all artifacts, and a fourth regenerates every number in the manuscript.

3 Against a null, the metrics mostly agree

We build the null by permuting the secondary metric across models within each group, which destroys any association with the primary metric while preserving both marginal distributions and all group sizes. Under independent rankings the expected flip rate is 0.5.

Table 1 gives the result. On TSB-AD-M the observed AUC-ROC-versus-Aff-F1 rate is 0.3145 against a null of 0.5002 (sd 0.0056, 200 permutations). On the 6-dataset grid the deep-model rate is $14/60 = 0.2333$ against 0.5004, and the seven-model rate is $44/126 = 0.3492$ against 0.5021. Every observed rate is far below its null.

The consequence is a reframing rather than a refutation. "65.1% of pairwise rankings are preserved across a point metric and a segment metric" and "0.3492 of pairwise rankings flip" are the same measurement. The first is the honest summary; the second, which is how such numbers are presented, invites the reader to supply a reference point of zero.

We claim no novelty for this step. Kendiukhov (2026) applies the same permutation construction to ranking reversals in gene regulatory network benchmarking and reaches the analogous conclusion, that a headline reversal rate of 0.163 against a null of 0.500 indicates substantial shared ranking structure. Within TSAD,

Table 1: Rank-flip rates against a random-ranking null. Reported rates are routinely discussed against an implicit reference of zero; the informative reference is the null column. All rates are AUC-ROC versus Aff-F1.

Grid	Models	Flips / pairs	Rate	Null	Agreement
TSB-AD-M	25	15,151/48,180	0.3145	0.5002	68.6%
TAB	5 deep	14/60	0.2333	0.5004	76.7%
TAB	7 (with classical)	44/126	0.3492	0.5021	65.1%

chance-corrected rank agreement is also already available in the form of Kendall's τ (Lyu, 2026). The point is that the raw percentage is what circulates, and it circulates without its baseline.

4 Most of the disagreement is a margin artifact

Two models separated by 0.001 AUC-ROC are not meaningfully ordered by AUC-ROC, so a metric that orders them differently has not reversed a finding. Because such pairs are numerous, they can carry a flip rate on their own.

Table 2 stratifies the TSB-AD-M pairs by $|\Delta\text{AUC-ROC}|$. The pattern is steep, and monotone above a margin of 0.02: pairs separated by less than 0.01 flip at 0.4743, essentially the chance rate, while pairs separated by at least 0.20 flip at 0.2015. Excluding every pair within 0.05 AUC-ROC retains 70.2% of pairs and gives 0.2556.

One-sided conditioning is not enough. It is tempting to stop here and report that roughly one pair in five still reverses at a margin no one would call ambiguous. That reading is wrong, and the error is instructive. Conditioning on $|\Delta\text{AUC-ROC}|$ selects pairs that the *primary* metric separates confidently while saying nothing about whether Aff-F1 separates them at all, so the surviving flips concentrate where the secondary margin is negligible. The symmetric stratification confirms the asymmetry: pairs with $|\Delta\text{Aff-F1}| < 0.02$ flip at 0.4999—the chance rate—and pairs with $|\Delta\text{Aff-F1}| \geq 0.40$ flip at 0.0848.

Requiring both metrics to separate the pair gives the two-sided statistic (Table 3). Among the 13.3% of pairs that *both* metrics separate by at least 0.20, only 0.0220 are ordered differently. The corresponding figures at joint margins of 0.05 and 0.10 are 0.1306 and 0.0651.

We regard the two-sided number as the closer answer to the question the literature is actually asking. "How often do two metrics confidently disagree about which of two detectors is better?" is not 0.3145, and it is not 0.2015. On this grid it is about two percent, against a random-ranking null of 0.4991 and with a cluster bootstrap interval of $[0.0092, 0.0417]$ over collections. It is also extremely unevenly distributed: 9 of the 16 collections that retain any doubly-separated pair contain *no* confident disagreement at all.

The concentration is worth stating on its own. At a joint margin of 0.20, confident disagreement is not a property of the benchmark suite but of two collections: SMAP (49/675 = 0.0726) and MSL (29/492 = 0.0589), both spacecraft telemetry. The clearest illustration of the argument of Section 5 is GHL, which has the highest one-sided pooled rate of any collection at 0.5876 and yet contains 0 confident disagreements in 63 doubly-separated pairs. The collection that looks worst under the contaminated statistic is one of the 9 that are perfectly clean under the corrected one.

Two properties of this estimand should be stated rather than left to the reader. Requiring both metrics to separate a pair conditions on the secondary metric's margin, which is not independent of sign agreement, so the two-sided rate is conservative in the same structural way the one-sided rate is anti-conservative; the two bracket the quantity of interest rather than either being unbiased. And the same absolute threshold is applied to two metrics with different dispersions, which makes the Aff-F1 condition the binding one: at 0.20 it cuts the $|\Delta\text{AUC-ROC}| \geq 0.20$ set from 16,045 pairs to 6,412. We therefore report both statistics throughout. The one-sided rate remains the right answer to a different and legitimate question—how often a reader who ranks by AUC-ROC would be contradicted by Aff-F1—and we do not claim it is meaningless, only that it is not a measure of confident disagreement.

Table 2: TSB-AD-M flip rate by AUC-ROC margin. Disagreement falls steeply with margin, but conditioning on the primary metric alone is one-sided; see Table 3.

$\lvert\Delta$AUC-ROC$\rvert$	Pairs	Share	Flip rate
< 0.01	4,700	9.8%	0.4743
0.01–0.02	3,139	6.5%	0.4839
0.02–0.05	6,513	13.5%	0.4230
0.05–0.10	7,757	16.1%	0.3616
0.10–0.20	10,026	20.8%	0.2603
≥ 0.20	16,045	33.3%	0.2015

Table 3: Two-sided margin conditioning on TSB-AD-M. Left: stratified by the secondary metric's margin. Right: both margins required to exceed a threshold. Confident disagreement is rare.

$\lvert\Delta$Aff-F1$\rvert$	Pairs	Flip	Both margins $\geq$	Pairs	Flip
< 0.02	12,883	0.4999	0.05	20,214	0.1306
0.02–0.05	8,601	0.3675	0.10	12,796	0.0651
0.05–0.10	7,404	0.3052	0.20	6,412	0.0220
0.10–0.20	7,628	0.2272			
0.20–0.40	6,417	0.1733			
≥ 0.40	5,247	0.0848			

A related concern is that flips are produced by detectors that do not discriminate at all. On the 6-dataset grid we drop every model–dataset row with $\lvert\text{AUC-ROC} - 0.5\rvert < \delta$ and recompute. The rate is 0.3492 at $\delta = 0$, then 0.3200, 0.3774, and $13/35 = 0.3714$ at $\delta = 0.02, 0.05, 0.10$. Removing uninformative detectors does not remove the disagreement.

Two cautions about this check. The surviving sample is small—35 pairs at $\delta = 0.10$—and the clustered interval is correspondingly wide, $[0.1765, 0.6155]$. And excluding by measured discriminative power is not the same as excluding by model family: Isolation Forest is the strongest detector by AUC-ROC on four of the six datasets, while Anomaly Transformer sits within 0.03 of chance on all six, so the $\delta = 0.10$ exclusion drops 16 deep rows against 5 classical ones. Pairs that mix a deep and a classical detector do flip more often ($27/60 = 0.4500$, versus $14/60 = 0.2333$ for deep–deep pairs), but this traces to Isolation Forest, not to classical methods as a class: split by which classical detector the pair contains, the rate is $16/30 = 0.5333$ for deep–IForest against $11/30 = 0.3667$ for deep–LOF. Across all 126 pairs Isolation Forest is involved in more flips than any other detector ($19/36 = 0.5278$), which is the same detector that leads on AUC-ROC on four of the six datasets. Being the strongest detector by one metric and the largest single source of disagreement is not a contradiction; it is what metric disagreement looks like when it is concentrated in one model.

5 Evaluation series are not independent

TSB-AD-M draws its 180 series from 17 source collections. Pooled flip rates differ sharply across them, from 0.1311 on TAO to 0.5876 on GHL. Treating series as independent replicates therefore overstates precision, and—more seriously—creates a confound for any covariate that is a property of the collection.

The short-anomaly ratio is such a covariate. Table 4 shows that α is constant within 11 of the 12 collections that contain more than one series, and within 7 of the 8 that contribute at least eight. It is, operationally, a collection label. Mean segment duration and segment count are not: both vary within every multi-series collection.

The consequences are in Table 5. At the series level α correlates with the flip rate at $\rho = 0.3241$ with a permutation p of 0.0002 against a resolution floor of 10^{-4}—an apparently decisive result. At the collection level the same association is 0.0563 with $p = 0.827$. Leave-one-collection-out is equally stark: removing TAO

Table 4: Within-collection variation of structural covariates on TSB-AD-M. A covariate that is constant within collections cannot be identified separately from collection identity at the series level; its between-collection variation remains admissible evidence, but at $n = 17$ rather than $n = 180$. Five collections contribute a single series and are constant for every covariate by construction, which is why the middle column is the informative one.

Covariate	All (17)	Multi-series (≥ 2 series: 12)	Large (≥ 8 series: 8)
$\alpha = 1 - N_{\text{short}}/N_{\text{total}}$	16	11	7
Mean segment duration	5	0	0
Segment count	5	0	0

Table 5: Spearman correlation between structural covariates and the one-sided flip rate, at series level and with collections as the unit, plus the range of the series-level ρ under leave-one-collection-out. Permutation p-values are bounded below by $1/(n_{\text{perm}} + 1) = 10^{-4}$ and are upper bounds. "Worst drop" names the collection whose removal minimises $|\rho|$. Against this estimand α is the only covariate that fails; Table 6 repeats the exercise against the two-sided rate, where duration fails as well.

Covariate	Series ρ	Coll. ρ	Coll. p	LOCO range	Worst drop
α	0.3241	0.0563	0.827	$[0.0898, 0.3741]$	TAO
Mean segment duration	0.3717	0.3750	0.138	$[0.2408, 0.4814]$	GHL
Segment count	-0.3775	-0.4118	0.098	$[-0.4252, -0.2535]$	TAO
Anomaly density	-0.3601	-0.3333	0.195	$[-0.4088, -0.2144]$	GHL
Series length	0.2047	0.2696	0.289	$[0.0394, 0.3452]$	GHL

alone takes the series-level correlation to 0.0898. The $\alpha = 0$ stratum on this grid consists of eleven TAO series and one other, so the low-α regime is one collection wearing the clothes of twelve observations.

The same holds on the 6-dataset grid, independently. Only 2 of 6 datasets have $\alpha < 1$, so the entire structural claim rests on them. SWaT, MSL, SMAP and WADI all have $\alpha = 1$ and ten deep-model pairs each, yet produce 5, 0, 1 and 1 flips respectively. Within a single α regime the flip count ranges over the full width of the effect α is supposed to explain, so on this grid α cannot separate these datasets at all.

Finally, spread within the $\alpha = 1$ regime exceeds spread across α regimes: among collections with $\alpha = 1$ throughout, pooled flip rates run from 0.1733 (GECCO) to 0.5876 (GHL), a wider range than the α-stratified contrast itself.

Interval estimates. The same non-independence affects uncertainty. Resampling series gives a 95% interval of $[0.2979, 0.3312]$ for the TSB-AD-M flip rate; a cluster bootstrap over collections gives $[0.2695, 0.3715]$, about three times wider. On the 6-dataset grid the effect is severe enough to be disqualifying: the deep-model interval is $[0.1333, 0.3500]$ treating the 60 pairs as independent Bernoulli draws, but $[0.0667, 0.4167]$ clustering over the six datasets. A study of that size cannot resolve a flip rate to better than a range more than a third of the unit interval wide—compatible with near-agreement and with near-chance simultaneously—and should not be used to argue about its magnitude.

6 A worked failure case: interpolation metrics at their endpoints

One response to metric disagreement is to combine metrics. In earlier work we proposed such a composite, which we called SAEScore, with the structure weight α selecting the blend:

$$\text{SAEScore}(D, M) = (1 - \alpha(D))\,\text{AUC-ROC}(M) + \alpha(D)\,\text{Aff-F1}(M), \qquad \alpha(D) = 1 - \frac{N_{\text{short}}(D)}{N_{\text{total}}(D)}. \tag{1}$$

Table 6: Structural covariates against the one-sided pooled flip rate (as in Table 5) and against the two-sided rate at a joint margin of 0.05, which retains 161 of 180 series. Permutation p in parentheses. Only segment count survives both corrections.

	One-sided		Two-sided (≥ 0.05 both)	
Covariate	Series ρ	Coll. ρ	Series ρ	Coll. ρ
α	0.3241	0.0563 (0.827)	0.0858 (0.291)	0.1279 (0.631)
Mean segment duration	0.3717	0.3750 (0.138)	-0.0056 (0.946)	0.1667 (0.521)
Segment count	-0.3775	-0.4118 (0.098)	-0.3294 (0.000)	-0.6225 (0.008)

We then stratified the AUC-ROC-versus-SAEScore flip rate by α and reported the resulting gradient—0.0000, 0.1758, 0.3321 across $\alpha = 0$, $0 < \alpha < 1$, and $\alpha = 1$—as confirming a structural regime split.

It confirms nothing. Equation 1 makes SAEScore identically AUC-ROC at $\alpha = 0$ and identically Aff-F1 at $\alpha = 1$; on our artifact the maximum absolute differences are 0.0 and 0.0 respectively. The $\alpha = 0$ stratum therefore compares AUC-ROC with itself and contributes 0 flips out of 3,594 by definition, and the $\alpha = 1$ stratum is bit-identical to the AUC-ROC-versus-Aff-F1 comparison (13,349/40,200 in both). The gradient is linear interpolation between a self-comparison and a fixed comparison. No configuration of the data could have produced anything else.

This generalizes past our own metric. Any composite defined as a convex combination of two metrics, with a weight that reaches its endpoints on real data, will exhibit a monotone "regime" pattern when stratified by that weight, whether or not the underlying structural hypothesis is true. Such a stratification is a description of the metric's definition, not a finding about benchmarks. The diagnostic is cheap: check whether the extreme strata of the weight reduce the composite to one of its constituents, and if they do, report the constituent comparison instead.

Doing that here gives strata of 0.1466, 0.2897, and 0.3321. These are metric-independent and still monotone—but as Section 5 shows, the gradient is carried by collection identity, and α does not survive as a predictor.

7 What survives

The structural hypothesis is not dead, but less of it survives than the clustering correction alone suggests. Against the one-sided pooled rate, two covariates hold up where α does not: mean segment duration at 0.3717 (series) and 0.3750 (collection), stable within $[0.2408, 0.4814]$ under leave-one-collection-out, and segment count at -0.3775 and -0.4118. Neither is constant within the multi-series collections, so unlike α both are identifiable.

Applying our own margin correction one level deeper removes one of them. Table 6 recomputes the same associations against the two-sided flip rate—the estimand Section 4 argues for—at a joint margin of 0.05, which retains 161 of 180 series. Mean segment duration goes to -0.0056 ($p = 0.946$) at the series level and 0.1667 ($p = 0.521$) at the collection level: it does not survive. It was tracking the contaminated part of the statistic. Segment count instead strengthens, to -0.3294 and -0.6225 with $p = 0.008$—the only collection-level association anywhere in this paper that is significant at conventional levels. α remains absent at 0.1279 ($p = 0.631$).

We report this because it is what our own method demands, and we note that it costs us one of two positive findings. The surviving statement is single and narrow: among the structural covariates we can measure, the number of anomaly segments in a series is the one that still tracks metric disagreement after both the clustering and the margin corrections, and it does so in the direction that fewer, longer segments mean more disagreement.

We state the limits plainly. With 17 collections these tests are underpowered, and at a joint margin of 0.20 the retained sample is small enough that nothing reaches significance, segment count included (-0.2139,

$p = 0.425$). What we claim is the comparison, not the coefficients: under identical treatment α fails every test we apply, duration fails the margin test, and segment count fails none of them.

Note also that these covariates are not independent of one another. Mean segment duration is algebraically determined by anomaly density, series length and segment count, so a regression that conditions on all three cannot say anything further about duration. Reporting several one-at-a-time associations, as in Table 5, is the honest presentation at this sample size.

8 A reporting protocol

The following costs no additional experiments and would have caught every issue above.

1. **Report the null.** State the random-ranking baseline next to any metric-disagreement statistic. For pairwise flip rates it is 0.5; obtain it by permuting one metric across models within each evaluation group.
2. **Stratify by *both* margins.** Report the flip rate among pairs that both metrics separate by a stated threshold, and say what fraction of pairs that retains. Conditioning on the primary metric alone selects pairs whose secondary margin is free to be negligible, and leaves a residual that is mostly near-ties on the metric one did not condition on.
3. **Cluster at the source.** Use the source collection or dataset, not the series, as the resampling unit, and report the clustered interval. Where the two intervals differ materially, report both.
4. **Check identifiability before conditioning.** Before explaining disagreement with a covariate, check whether that covariate varies within clusters. If it does not, a series-level association is a between-collection association with an inflated n.
5. **Leave one collection out.** Report the range of any structural association under removal of each collection in turn. A result carried by one collection should be presented as such.
6. **Do not stratify a composite by its own weight.** If a metric reduces to a constituent at the extremes of a weight, stratifying by that weight measures the definition.

9 Related work

Our object of study is a statistic, not a metric, so the closest work is methodological rather than metric-proposing.

Critiques of TSAD evaluation. Kim et al. (2022) showed that point-adjustment lets random scores outperform published detectors, which is the canonical demonstration that a TSAD metric can be uninformative. Wu & Keogh (2023) and Liu & Paparrizos (2024) argue that benchmark datasets themselves are flawed, mislabeled, or trivially solvable, and Schmidl et al. (2022), Blázquez-García et al. (2021) and Boniol et al. (2024) survey the resulting landscape. Lyu (2026) stress-tests post-point-adjustment replacement metrics and finds that several, including affiliation-based and ROC-based scores, inflate steeply under best-of-N seed selection while PR-based metrics stay flat. That result is complementary to ours and sharpens it: the metric on one side of our comparison has a known instability under selection, which is a further reason not to read a raw flip count as a property of the benchmark.

Replacement metrics. Range-based precision and recall (Tatbul et al., 2018), affiliation metrics (Huet et al., 2022), VUS (Paparrizos et al., 2022), PATE (Ghorbani et al., 2024), DQE (Li et al., 2026), and confidence-consistency evaluation (Zhong et al., 2025) each repair a specific deficiency of point-wise scoring. Wagner et al. (2025) approaches the space axiomatically, specifying properties a metric ought to satisfy, and Yang et al. (2025) organize the resulting metric space by the evaluation problem each metric targets—including a dimension for robustness against random-score inflation, which is chance baselining applied to metric values. Our contribution is orthogonal to all of these: whichever metric one adopts, comparisons *between* metrics are currently reported without the ingredients above.

Ranking instability as a statistic. Quantifying how far rankings move under a change of protocol is now common practice across ML evaluation, from preference-data perturbations in language model leaderboards (Huang et al., 2025) to protocol axes in computational biology (Kendiukhov, 2026). The latter is the closest methodological neighbour to our Section 3: it permutes method scores within each condition over 5,000 replicates and reports an observed reversal rate of 0.163 against a null mean of 0.500, reaching the same conclusion in a different domain. It does not take up the question of Section 5 — its rate estimates use Wilson binomial intervals, which treat the constituent pairs as independent draws.

Statistical practice in benchmark evaluation. The concern that evaluation units are not independent replicates is old outside machine learning (Hurlbert, 1984). Within it, the argument that point estimates on small benchmark suites mislead without interval estimates has been made forcefully in reinforcement learning (Agarwal et al., 2021) and for language model evaluations (Miller, 2024), and Röchner et al. (2025) argue on general grounds that anomaly detection benchmarks should be organized into scenario groups sharing relevant characteristics. Ours is the quantitative counterpart to that last argument: grouping is not only a taxonomic convenience but the difference between a structural association of 0.3241 and one of 0.0563. Relative to Agarwal et al. (2021) and Miller (2024), which resample runs or items as exchangeable units, our point is that the exchangeable unit here is the source collection, and that identifying it changes which covariates remain admissible at all.

10 Limitations

Our conclusions are conditioned on a specific metric pair, AUC-ROC versus Aff-F1 at cardinality `one` with contamination-rate thresholding. Other pairs may behave differently, and the thresholding rule is known to matter (Masakuna et al., 2024). The TSB-AD-M slice follows file-ID order and is model-coverage-limited rather than a complete benchmark run, so it should be read as a replication of the statistical phenomenon and not as a TSB-AD leaderboard claim. With 17 collections and 6 datasets, the cluster-level analyses that carry our main argument are themselves underpowered—which is part of the point, but does mean our surviving positive claim about segment count is directional rather than established. We have not shown that any published conclusion in the TSAD literature is wrong as a result of these omissions; we have shown that the evidence offered for one class of conclusion, including ours, does not have the strength attributed to it.

11 Conclusion

A rank-flip rate is a compact and persuasive statistic, and it is currently reported in a way that cannot be interpreted. Supplied with a null, the flip rates in this literature and in our own prior work describe metrics that substantially agree. Stratified by both margins, most of the apparent residual turns out to be near-ties on the metric one did not condition on. Clustered at the source collection, the structural account of that disagreement does not survive, and neither does the composite metric we built on it. What we can defend is narrower than what we previously claimed, and we think it is more useful: on TSB-AD-M, point and segment metrics disagree on about two percent of the pairs that both of them clearly separate, what disagreement there is varies across benchmark collections to a degree that dwarfs any anomaly-structure effect we can identify, and the one structural covariate that still tracks it under both corrections is a continuous one the field has not been using.

Broader Impact Statement

This work audits an evaluation statistic and proposes reporting practice; it introduces no model, no dataset, and no deployable artifact. The direct effect we anticipate is that some published claims about metric instability in time-series anomaly detection will be restated more conservatively, including our own. We see one indirect risk worth naming. Showing that a widely cited instability figure overstates disagreement could be read as licence to keep reporting single point-wise metrics on benchmarks whose anomalies are sustained segments. That reading would be wrong, though our results make the point less forcefully than the ones they replace: the labeling mismatch documented in prior work is untouched by anything we show, and a metric can be misaligned with a task even when it happens to rank detectors much as another metric does. The benchmarks we use are public industrial and infrastructure datasets containing no personal data.

Reproducibility statement

All results are recomputed from committed JSON artifacts by three scripts, and every numeric claim in this manuscript is generated into `numbers.tex` by a fourth, so the text cannot diverge from the analysis. The statistical helpers are implemented without a scipy dependency and are unit-tested against hand-computed values. The repository includes the derived metric rows, the taxonomy summary, and the reproduction commands; it does not redistribute the source benchmark datasets, for which access instructions are provided.

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